SCEC TPV102, strike slip rate at the nine on-fault stations (rows: depth 3 / 7.5 / 12 km; columns: strike)

flt_n12_3 (strike -12 km, depth 3 km)

flt_0_3 (strike 0 km, depth 3 km)

flt_12_3 (strike 12 km, depth 3 km)

flt_n9_7.5 (strike -9 km, depth 7.5 km)

flt_0_7.5 (strike 0 km, depth 7.5 km)

flt_9_7.5 (strike 9 km, depth 7.5 km)

flt_n12_12 (strike -12 km, depth 12 km)

flt_0_12 (strike 0 km, depth 12 km)

flt_12_12 (strike 12 km, depth 12 km)

strike slip rate (m/s)

time (s)

DRDG3D

PyLith

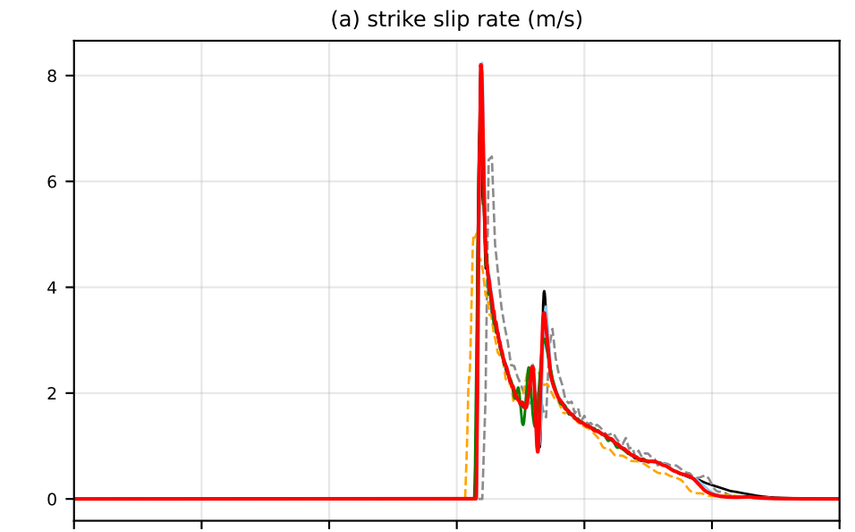
MFEM mixed-flux RK45 p=2 (gold)

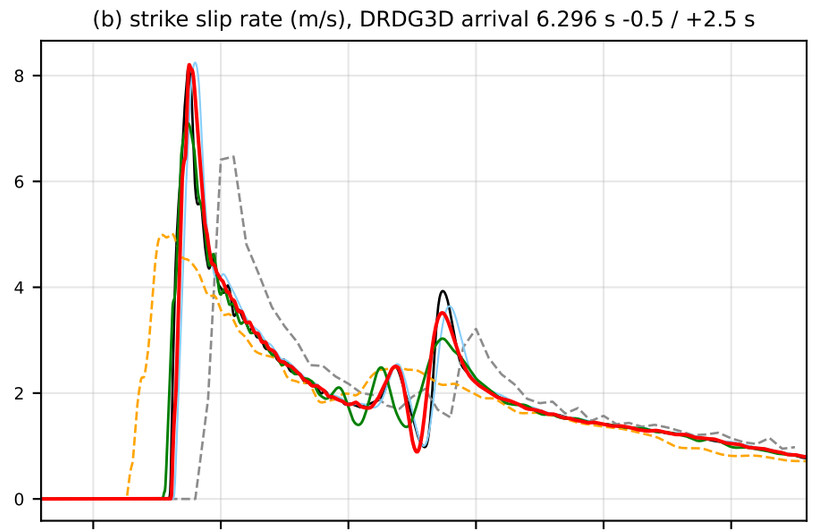
Nitsche damped p=1 200 m (Frontera 7995743)

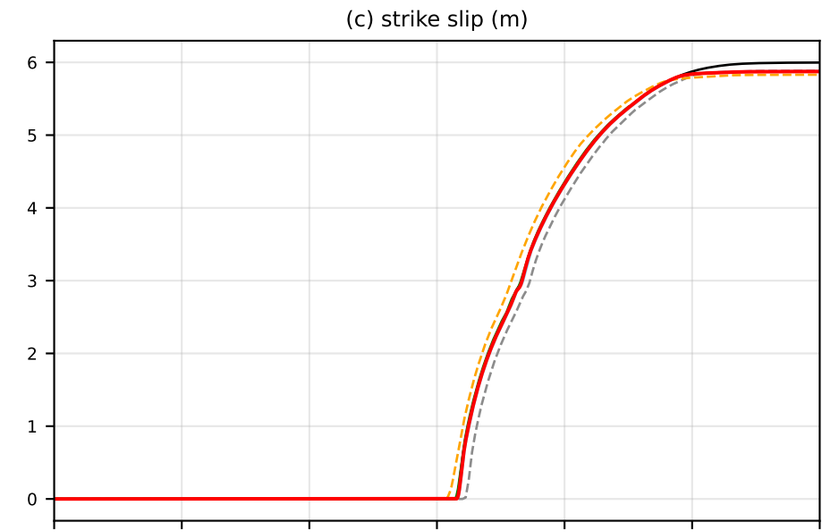
Nitsche damped p=2 200 m (Frontera 7995488, to t = 9 s)

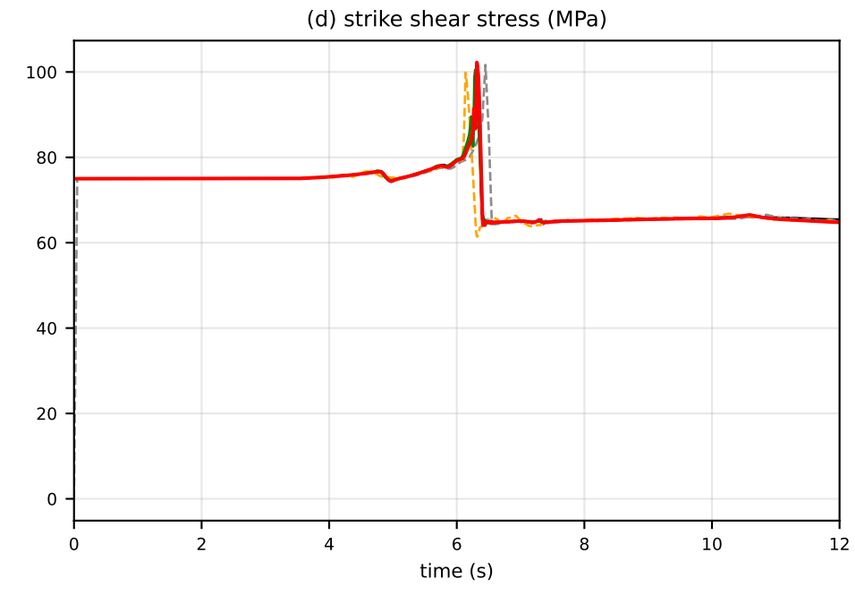
Nitsche p=3 200 m (Vista 1080363)

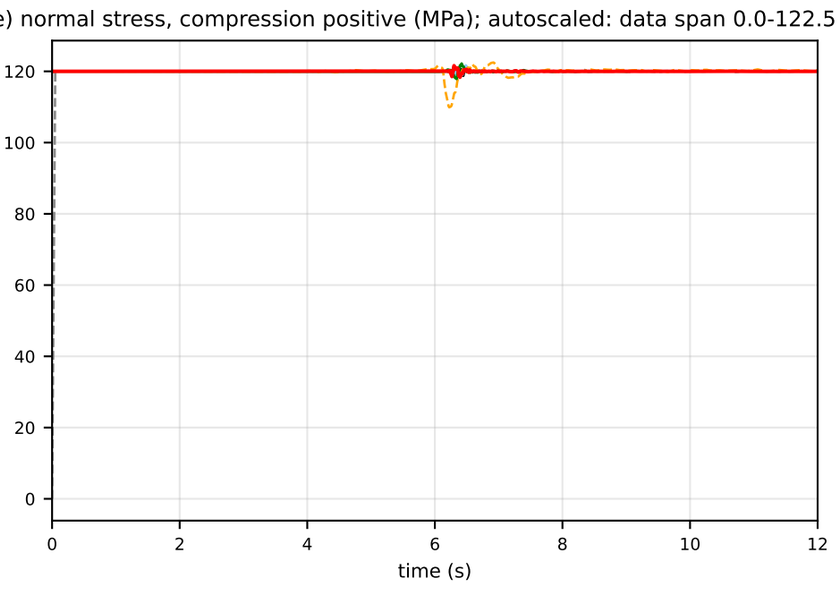
SCEC TPV102 station flt_n12_3 (strike -12 km, depth 3 km)
Nitsche p=3 200 m (Vista 1080363) vs DRDG3D: dt +0.007 s slip -2.05% peak +0.8%

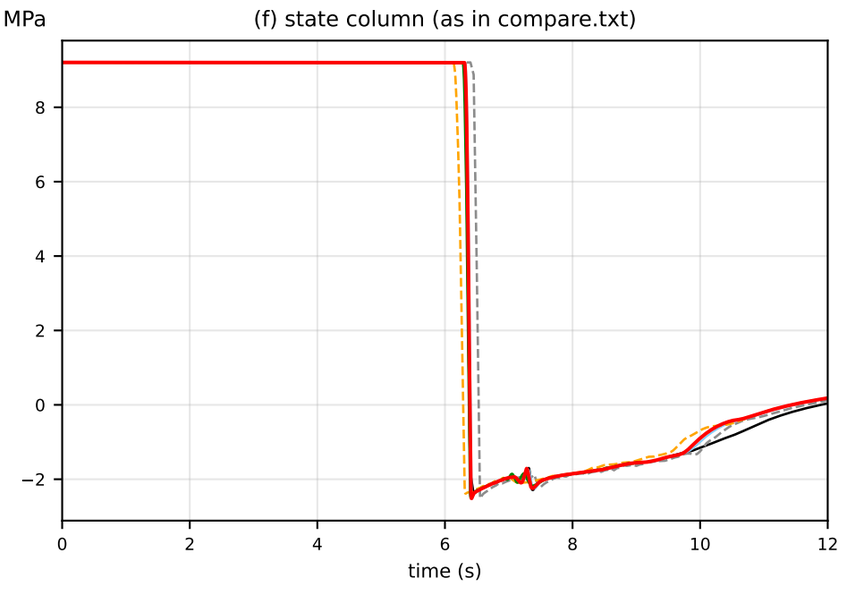






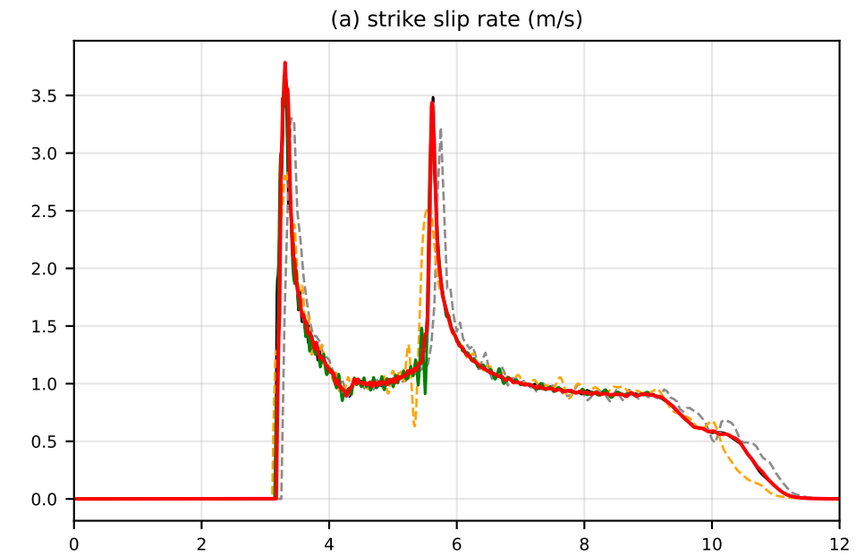


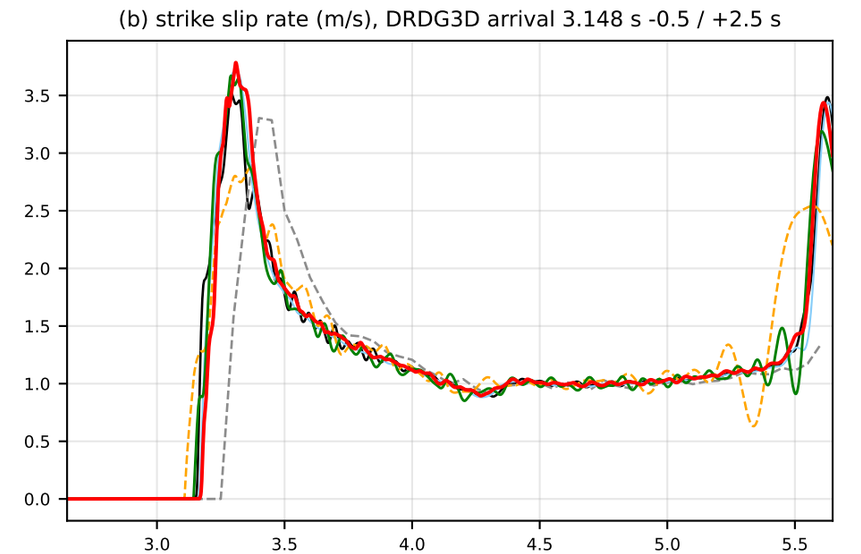




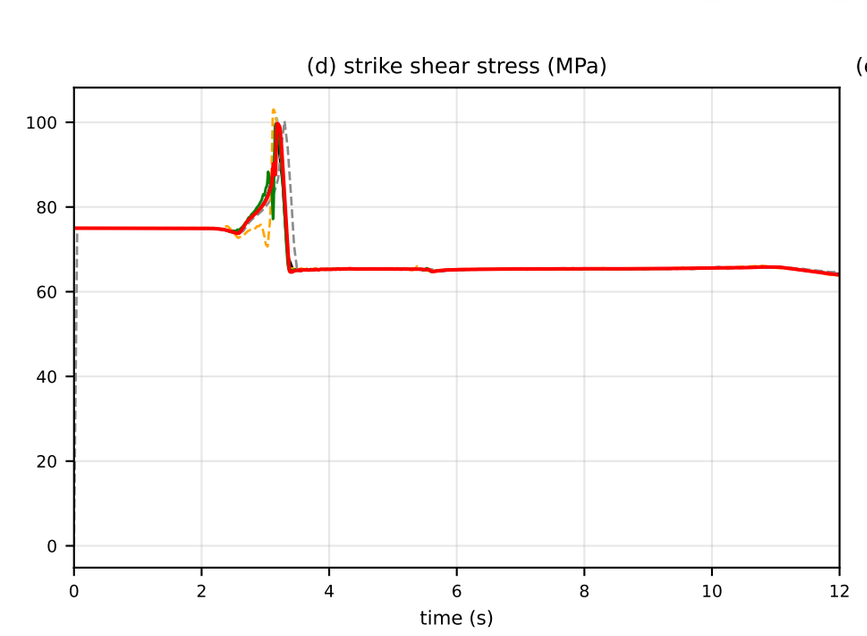
DRDG3D
PyLith
MFEM mixed-flux RK45 p=2 (gold)
Nitsche damped p=1 200 m (Frontera 7995743)
Nitsche damped p=2 200 m (Frontera 7995488, to t = 9 s)
Nitsche p=3 200 m (Vista 1080363)

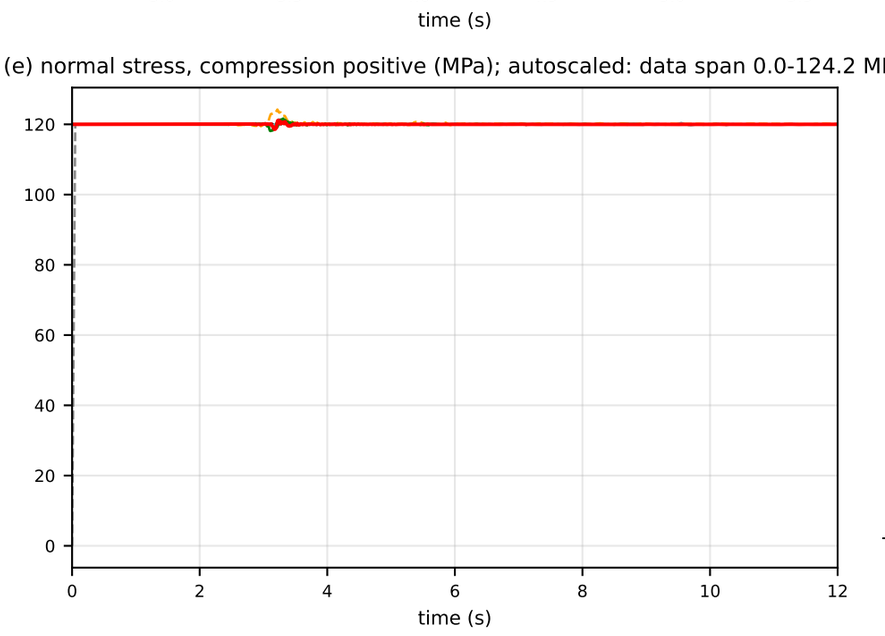
SCEC TPV102 station flt_0_3 (strike 0 km, depth 3 km)
Nitsche p=3 200 m (Vista 1080363) vs DRDG3D: dt +0.018 s slip -0.03% peak +7.3%

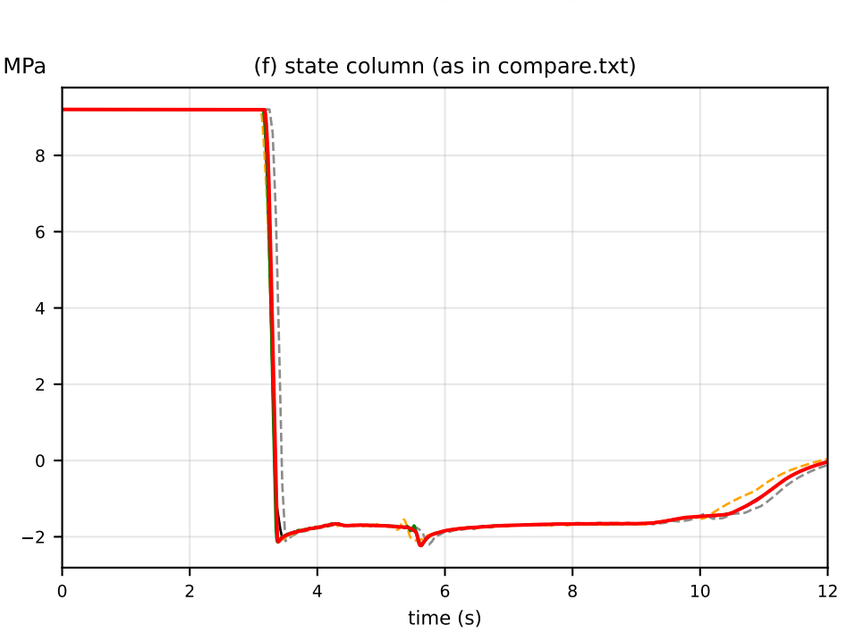


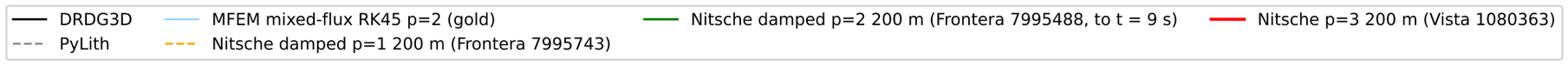






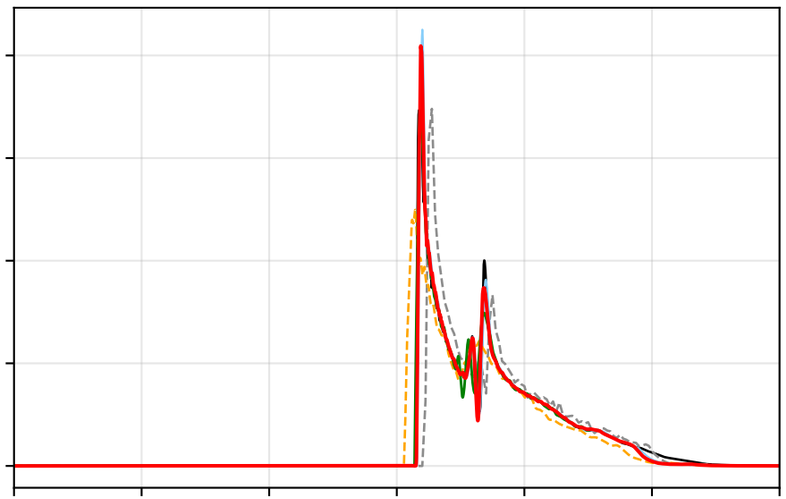




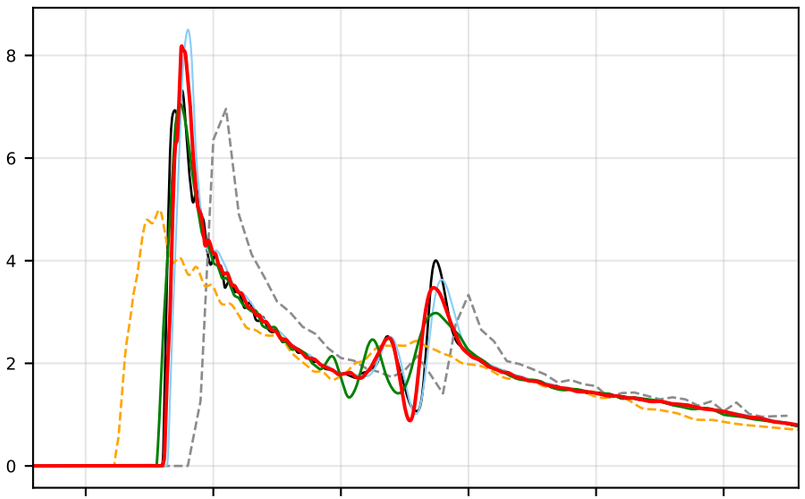


SCEC TPV102 station flt_12_3 (strike 12 km, depth 3 km)
Nitsche p=3 200 m (Vista 1080363) vs DRDG3D: dt -0.035 s slip -2.05% peak +11.9%

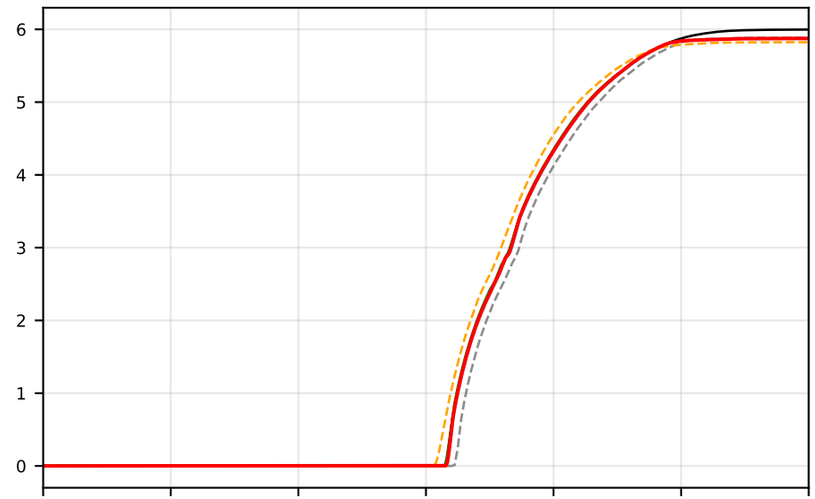
(a) strike slip rate (m/s)



(b) strike slip rate (m/s), DRDG3D arrival 6.294 s -0.5 / +2.5 s

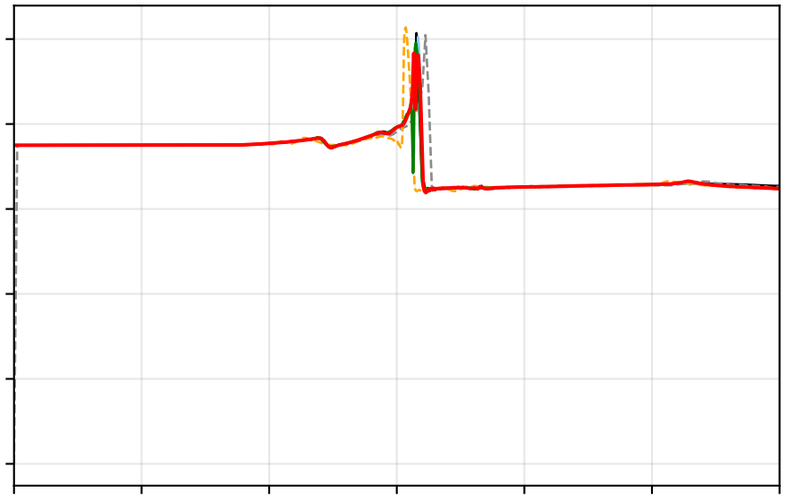


(c) strike slip (m)

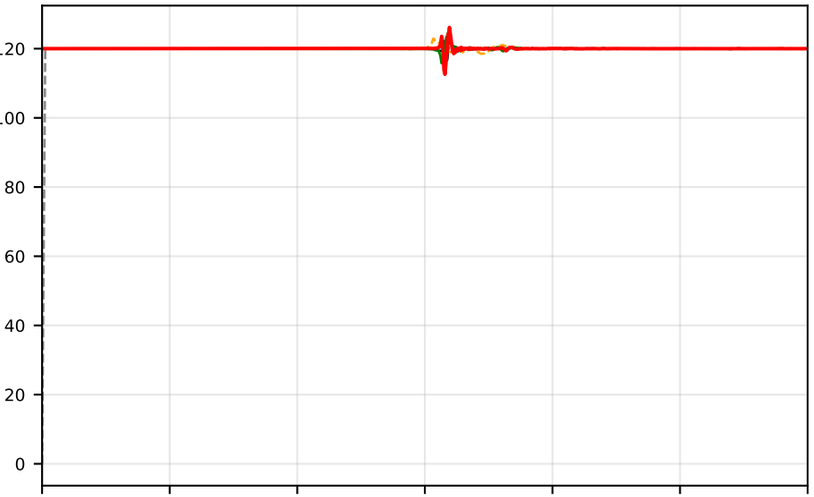


time (s)

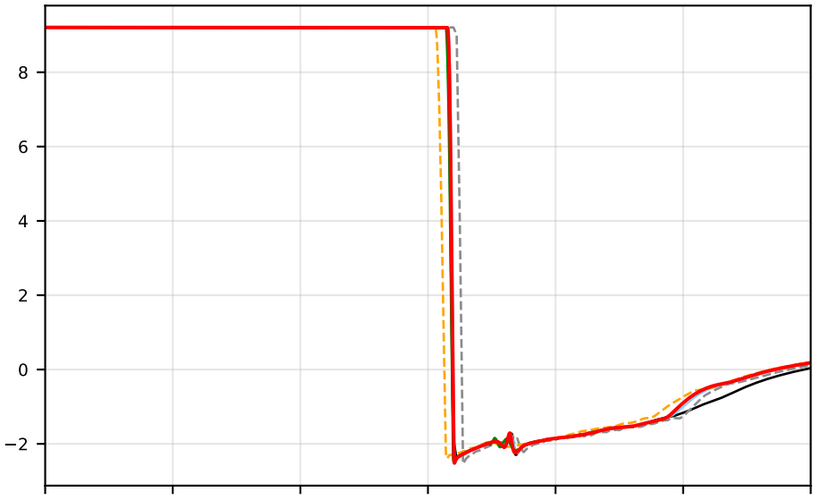
(d) strike shear stress (MPa)



(e) normal stress, compression positive (MPa); autoscaled: data span 0.0-126.1 MPa



(f) state column (as in compare.txt)



DRDG3D
PyLith
MFEM mixed-flux RK45 p=2 (gold)
Nitsche damped p=1 200 m (Frontera 7995743)
Nitsche damped p=2 200 m (Frontera 7995488, to t = 9 s)
Nitsche p=3 200 m (Vista 1080363)

SCEC TPV102 station flt_n9_7.5 (strike -9 km, depth 7.5 km)
Nitsche p=3 200 m (Vista 1080363) vs DRDG3D: dt +0.027 s slip -0.47% peak -0.9%

(a) strike slip rate (m/s)

(b) strike slip rate (m/s), DRDG3D arrival 4.713 s -0.5 / +2.5 s

(c) strike slip (m)

time (s)

(d) strike shear stress (MPa)

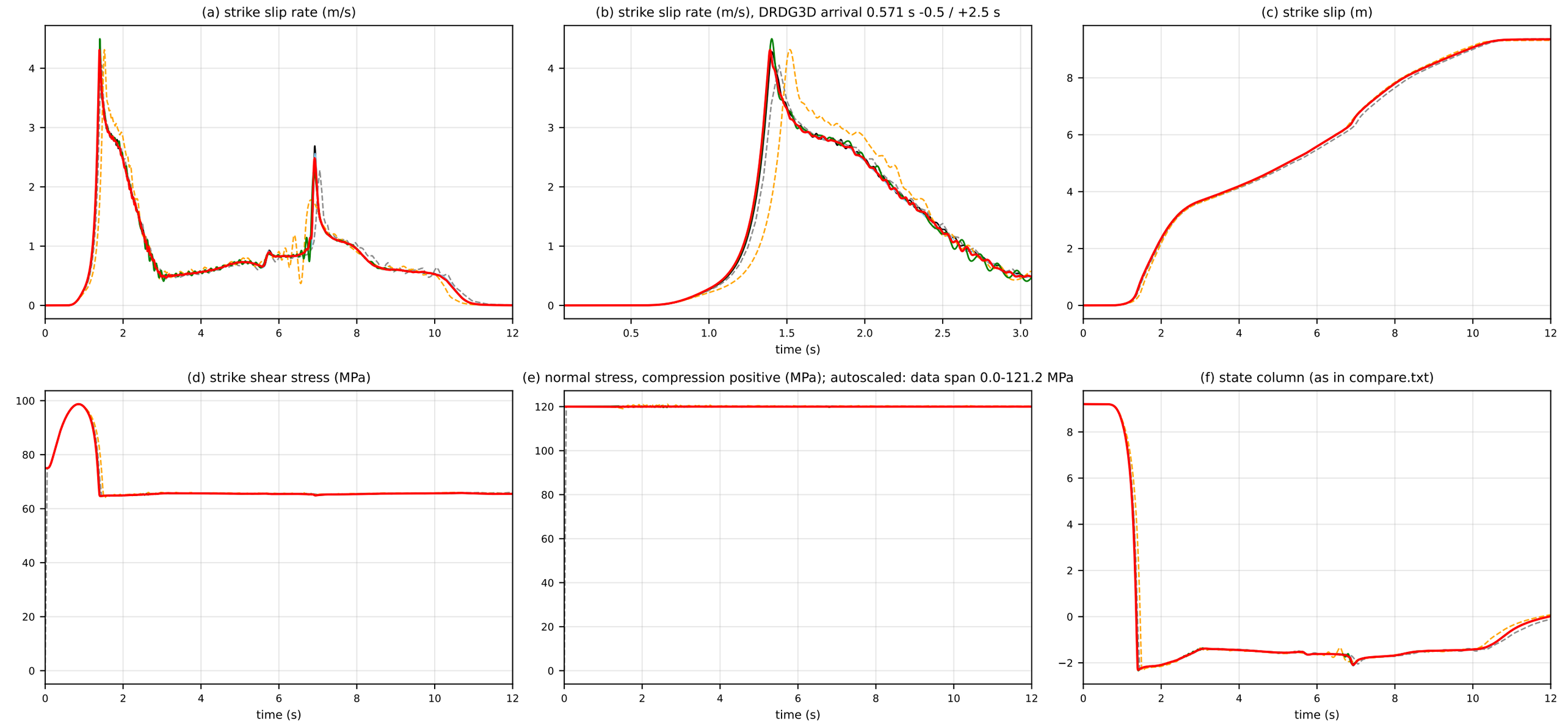
(e) normal stress, compression positive (MPa); autoscaled: data span 0.0-122.1 MPa

(f) state column (as in compare.txt)

time (s)

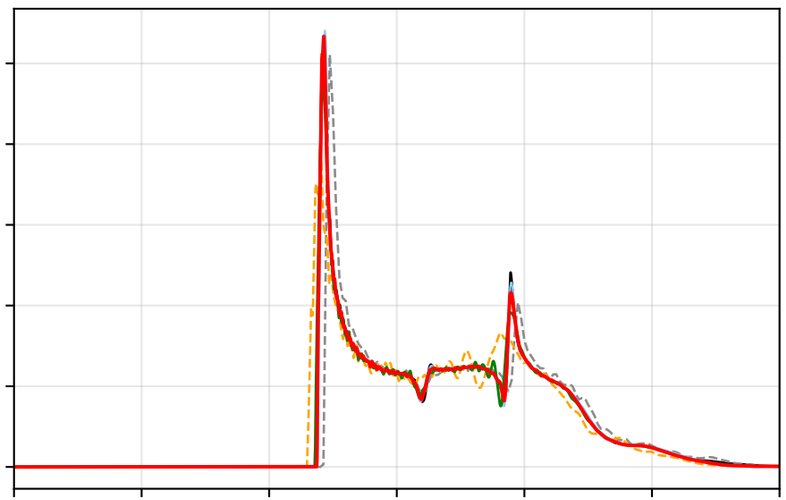
DRDG3D
PyLith
MFEM mixed-flux RK45 p=2 (gold)
Nitsche damped p=1 200 m (Frontera 7995743)
Nitsche damped p=2 200 m (Frontera 7995488, to t = 9 s)
Nitsche p=3 200 m (Vista 1080363)

SCEC TPV102 station flt_0_7.5 (strike 0 km, depth 7.5 km)
Nitsche p=3 200 m (Vista 1080363) vs DRDG3D: dt -0.002 s slip -0.02% peak +0.7%

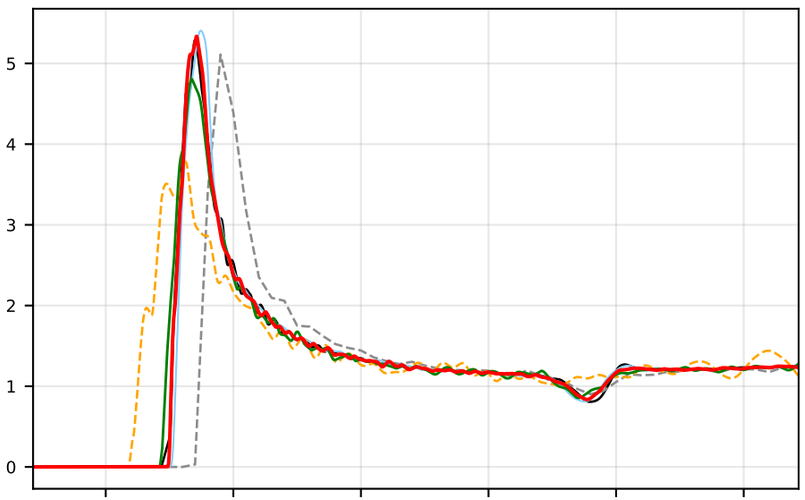


SCEC TPV102 station flt_9_7.5 (strike 9 km, depth 7.5 km)
Nitsche p=3 200 m (Vista 1080363) vs DRDG3D: dt +0.026 s slip -0.48% peak +1.0%

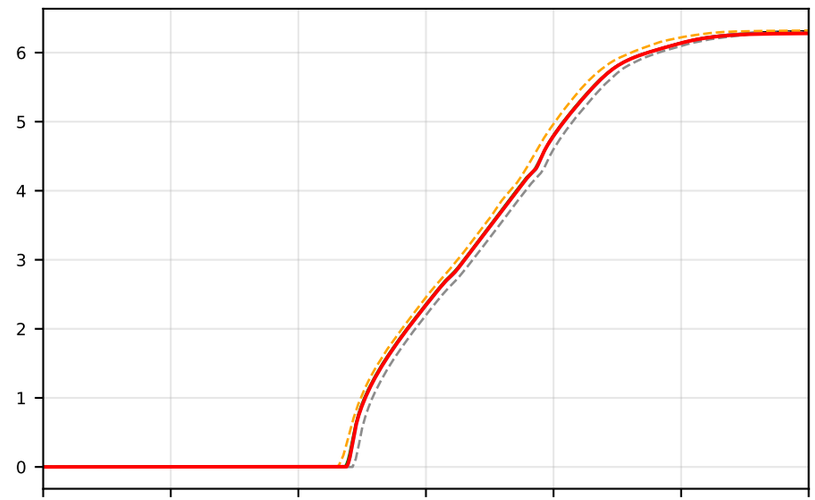
(a) strike slip rate (m/s)



(b) strike slip rate (m/s), DRDG3D arrival 4.715 s -0.5 / +2.5 s

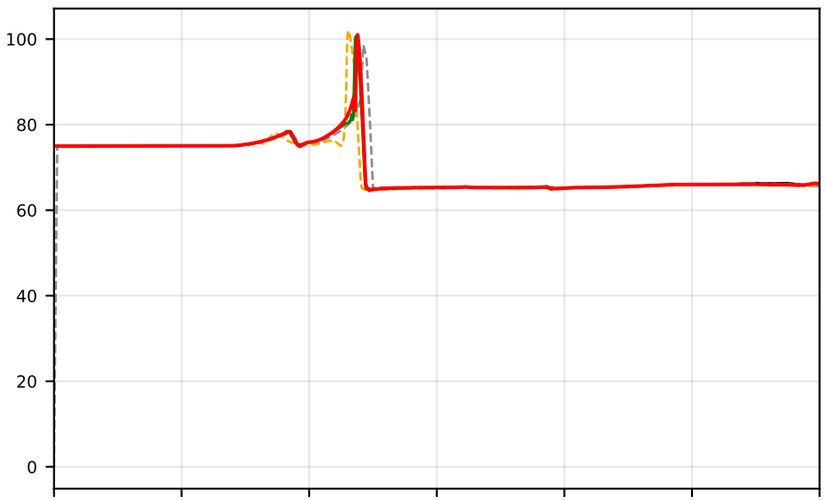


(c) strike slip (m)

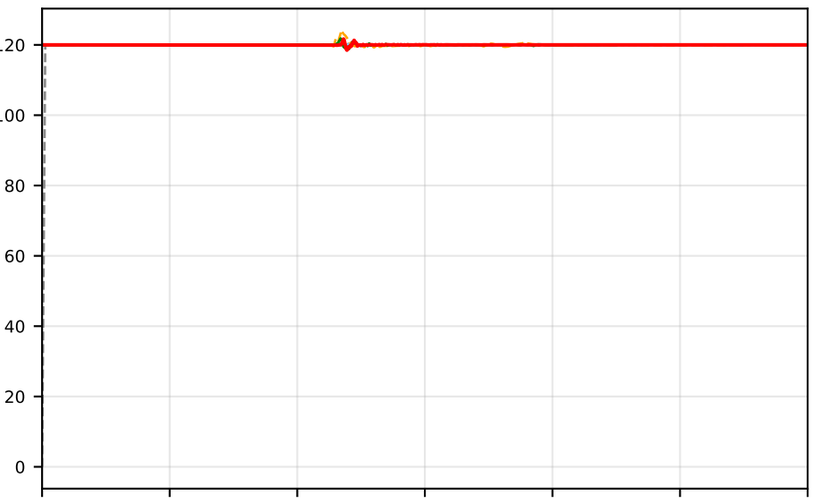


time (s)

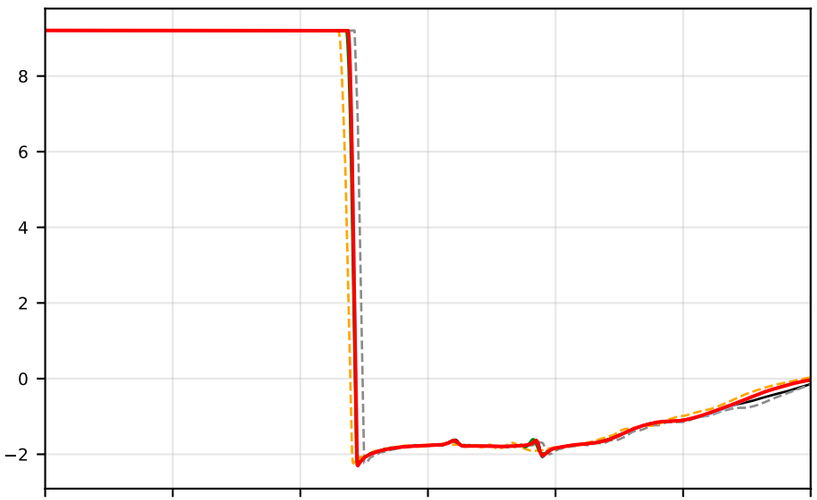
(d) strike shear stress (MPa)



(e) normal stress, compression positive (MPa); autoscaled: data span 0.0-124.1 MPa



(f) state column (as in compare.txt)



DRDG3D
PyLith
MFEM mixed-flux RK45 p=2 (gold)
Nitsche damped p=1 200 m (Frontera 7995743)
Nitsche damped p=2 200 m (Frontera 7995488, to t = 9 s)
Nitsche p=3 200 m (Vista 1080363)

SCEC TPV102 station flt_n12_12 (strike -12 km, depth 12 km)
Nitsche p=3 200 m (Vista 1080363) vs DRDG3D: dt +0.037 s slip +0.13% peak +2.7%

(a) strike slip rate (m/s)

(b) strike slip rate (m/s), DRDG3D arrival 6.247 s -0.5 / +2.5 s

(c) strike slip (m)

time (s)

(d) strike shear stress (MPa)

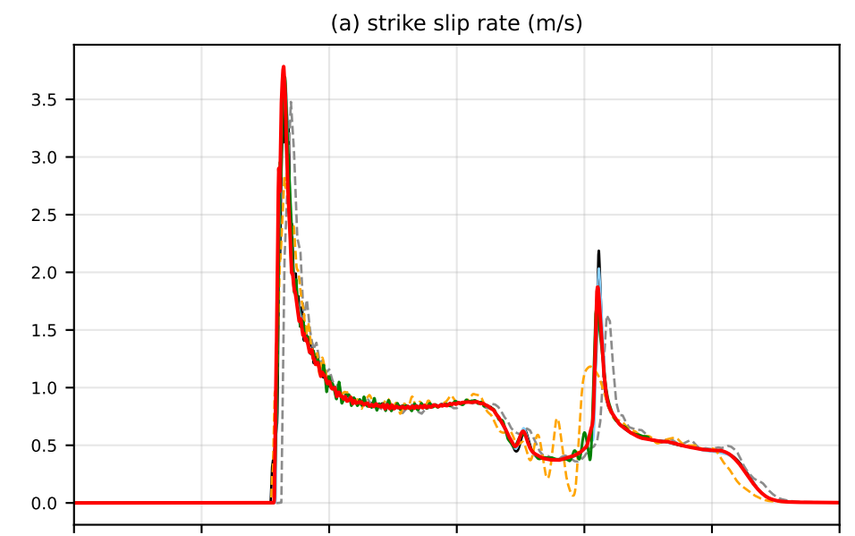
(e) normal stress, compression positive (MPa); autoscaled: data span 0.0-130.5 MPa

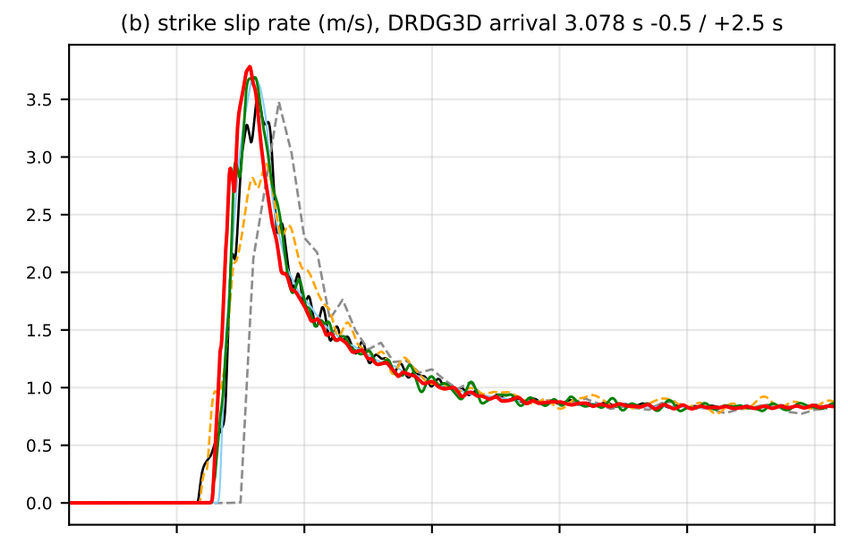
(f) state column (as in compare.txt)

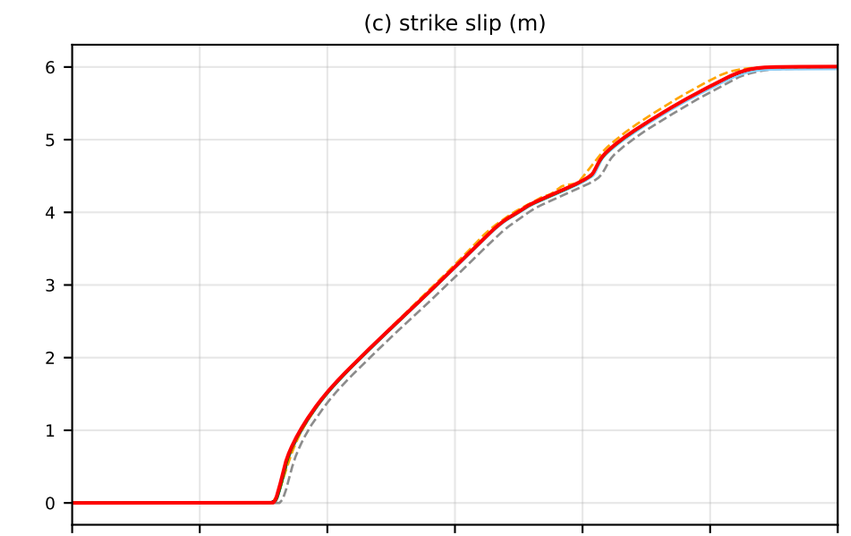
time (s)

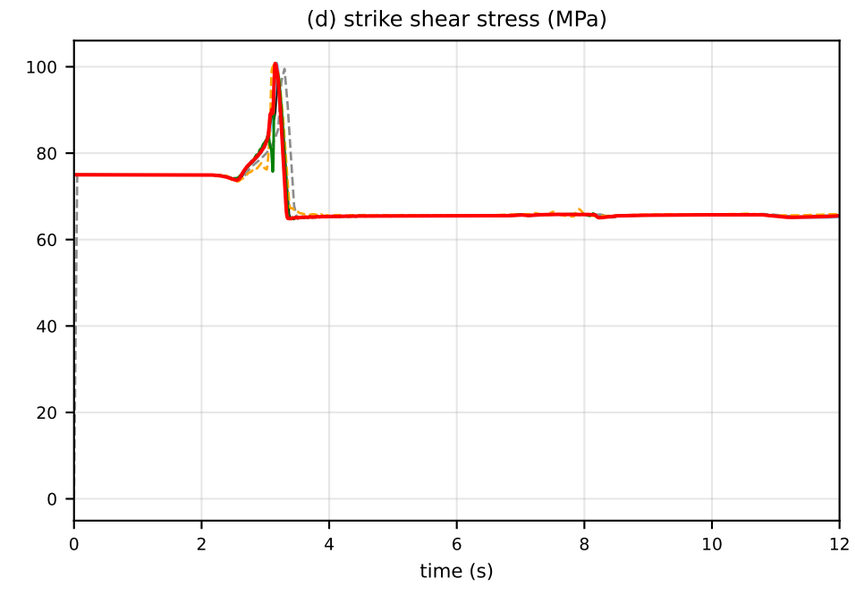
DRDG3D
PyLith
MFEM mixed-flux RK45 p=2 (gold)
Nitsche damped p=1 200 m (Frontera 7995743)
Nitsche damped p=2 200 m (Frontera 7995488, to t = 9 s)
Nitsche p=3 200 m (Vista 1080363)

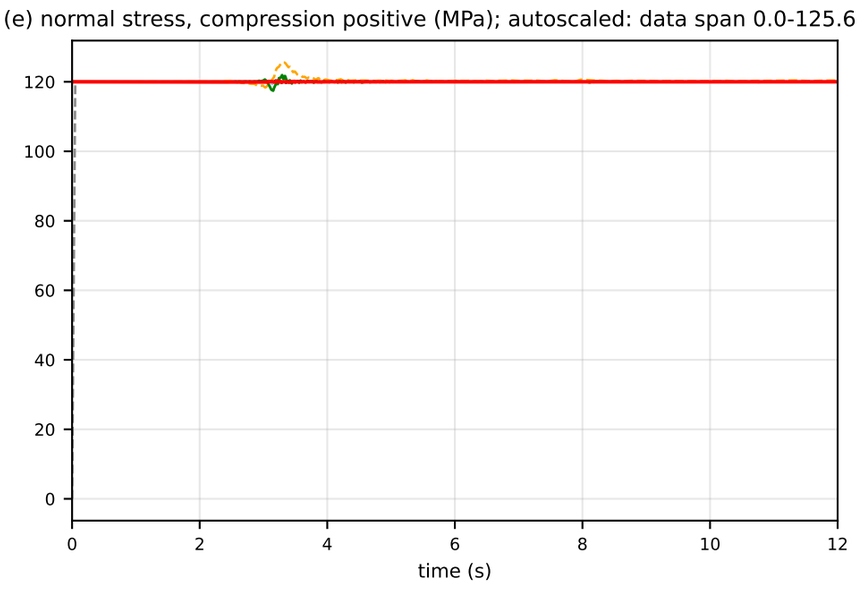
SCEC TPV102 station flt_0_12 (strike 0 km, depth 12 km)
Nitsche p=3 200 m (Vista 1080363) vs DRDG3D: dt +0.053 s slip +0.16% peak +7.0%

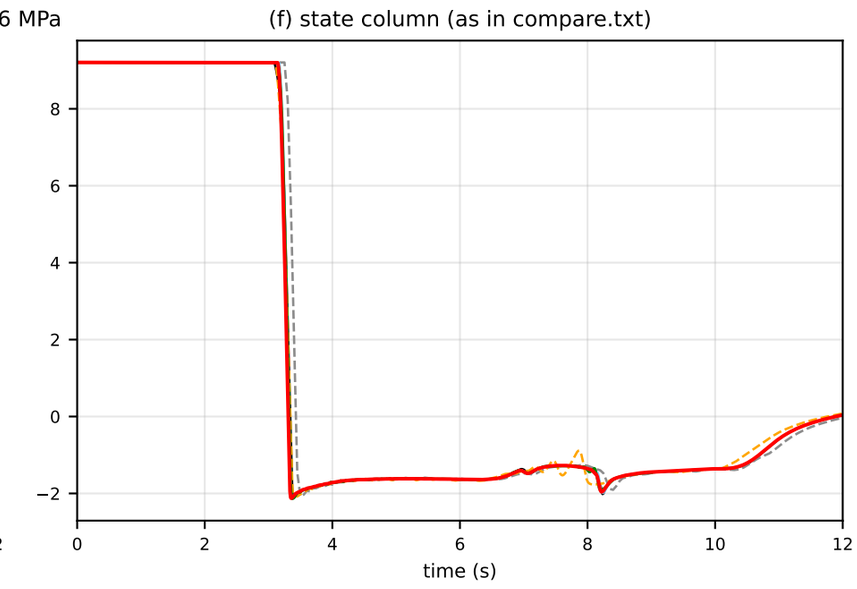


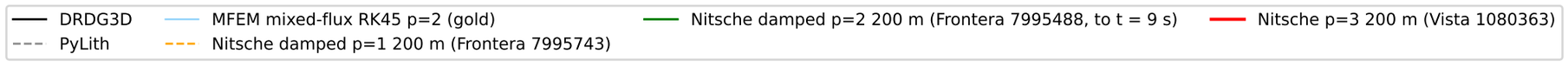




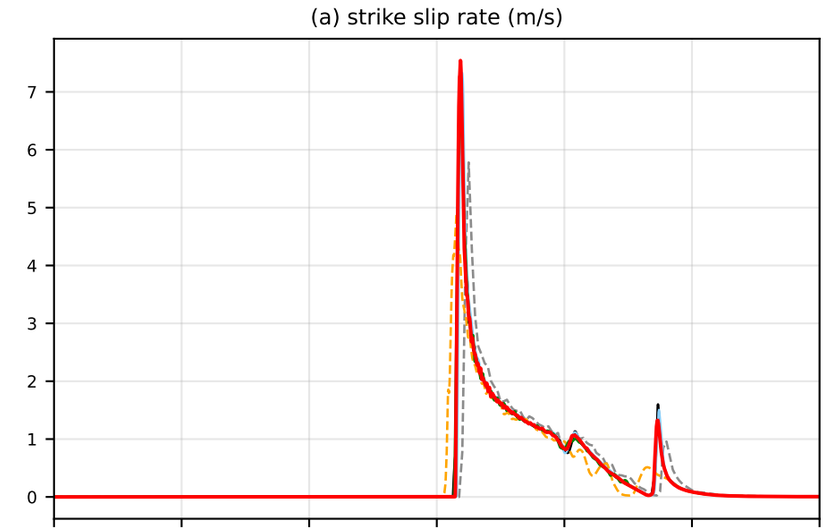


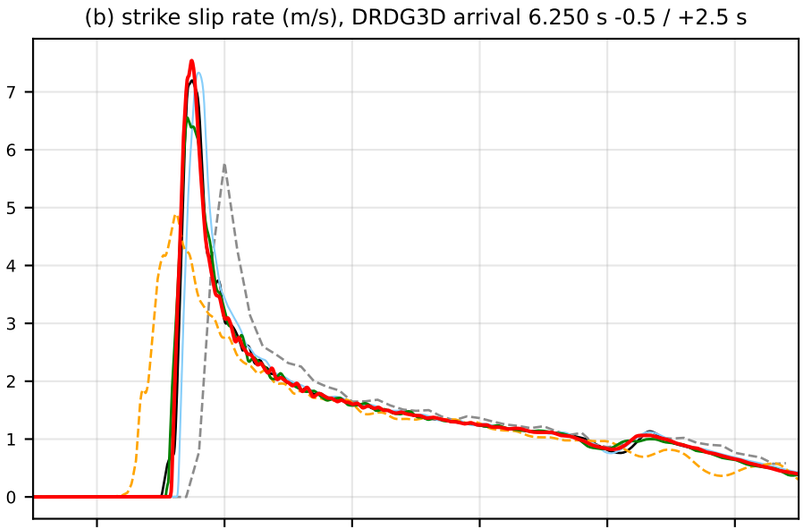






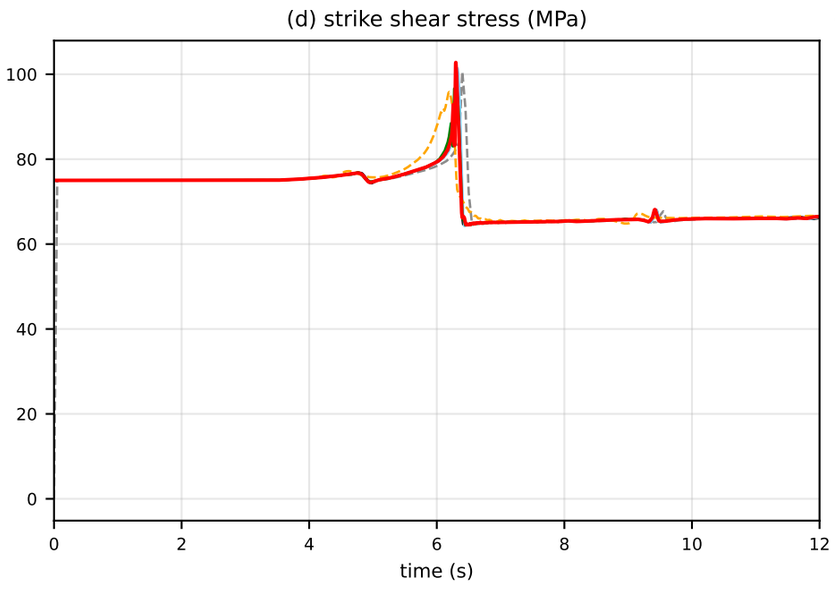
SCEC TPV102 station flt_12_12 (strike 12 km, depth 12 km)
Nitsche p=3 200 m (Vista 1080363) vs DRDG3D: dt +0.036 s slip +0.13% peak +4.8%

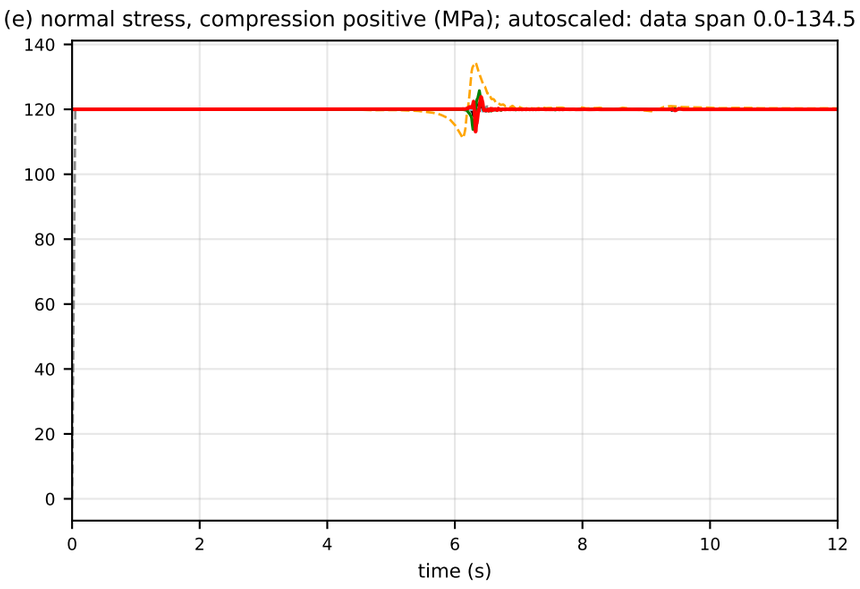


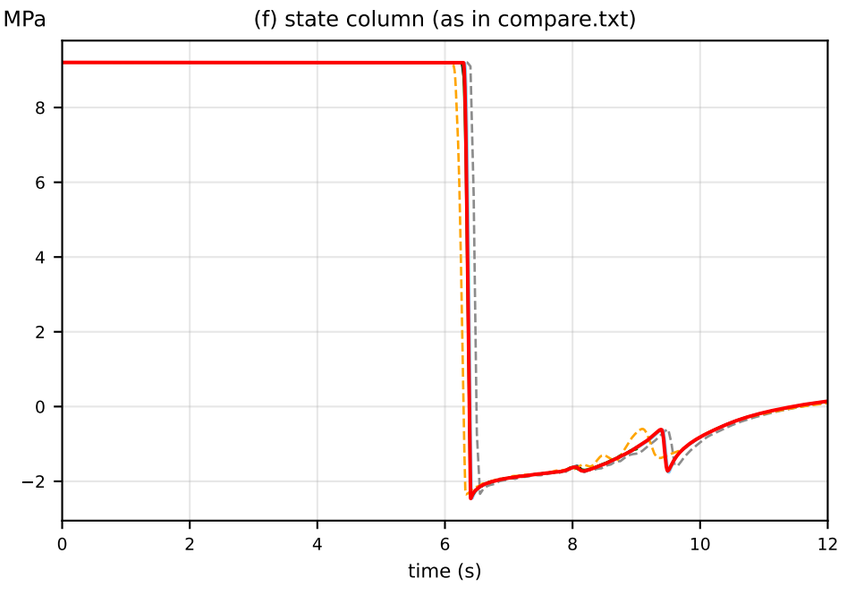




time (s)







DRDG3D
PyLith
MFEM mixed-flux RK45 p=2 (gold)
Nitsche damped p=1 200 m (Frontera 7995743)
Nitsche damped p=2 200 m (Frontera 7995488, to t = 9 s)
Nitsche p=3 200 m (Vista 1080363)